4.3.25 Earphones

These are small hemisphere-shaped headphones that can be positioned right in front of the auditory canal. They sometimes also have a retaining extension that goes around the ear that prevents them from falling off.

Supra-aural headphones

4.3.26 Noise Cancellation

Sound is in the form of vibrations. If two vibrations of the same magnitude but inverse directions collide, they nullify. This is the idea behind the concept of Noise Cancellation. Two type of Noise Cancellation techniques are used. Active Noise Cancellation uses microphones placed on the outside of the headphones to detect ambient noise. This information is used to create sound waves opposite to the ambient sound, thus nullifying the ambient noise. An additional power source is needed to drive the microphone, and this is usually in the form a batteries fitted into each ear-cup. Passive Noise cancellation relies on blocking out ambient noise by ensuring a tighter fit between the ear and the speaker. This technique is better implemented in circumaural and intra-aural headphones.

Earphones

4.3.27 Impedance

This is the load that a speaker puts on an amplifier. Speakers with low impedance can be operated without external power, while speakers with higher impedances need external power to work. Higher-impedance speakers produce louder, and typically better, sound.